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Valuing avoided morbidity using meta-regression analysis: what can health status measures and QALYs tell us about WTP?

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Summary

Many economists argue that willingness-to-pay (WTP) measures are most appropriate for assessing the welfare effects of health changes. Nevertheless, the health evaluation literature is still dominated by studies estimating nonmonetary health status measures (HSMs), which are often used to assess changes in quality-adjusted life years (QALYs). Using meta-regression analysis, this paper combines results from both WTP and HSM studies applied to acute morbidity, and it tests whether a systematic relationship exists between HSM and WTP estimates. We analyze over 230 WTP estimates from 17 different studies and find evidence that QALY-based estimates of illness severity – as measured by the Quality of Well-Being (QWB) Scale – are significant factors in explaining variation in WTP, as are changes in the duration of illness and the average income and age of the study populations. In addition, we test and reject the assumption of a constant WTP per QALY gain. We also demonstrate how the estimated meta-regression equations can serve as benefit transfer functions for policy analysis. By specifying the change in duration and severity of the acute illness and the characteristics of the affected population, we apply the regression functions to predict average WTP per case avoided. Copyright © 2006 John Wiley & Sons, Ltd.

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Introduction

Economic evaluation methods – in particular cost–benefit analysis (CBA) and cost-effectiveness analysis (CEA) – are now widely used to analyze and set priorities for public health and safety programs. Whereas CEA continues to be the method of choice for evaluating health care programs [1], CBA is often required by governments for evaluating environmental, food safety, and other regulatory programs.

A primary difference between these two methods is the metric they use to evaluate changes in health risks or outcomes. CBA requires that health changes be expressed in monetary terms. Although placing a dollar value on human health is often controversial, welfare economists generally agree that willingness-to-pay (WTP) provides a conceptually appropriate monetary measure of health benefits. In contrast, CEA stops short of monetizing health impacts and instead relies a wide variety of health-based effectiveness measures. Quality-adjusted life years (QALYs) in particular are now commonly used as a nonmonetary summary measure of effectiveness [2].

Because of CBAs prominent role in evaluating public health and safety policies, it is essential for

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policy analysts in these areas to be able to consistently estimate WTP for avoided illness. However, faced with inevitable time and resource constraints, analysts are typically limited in their ability to conduct primary health valuation research for specific policy initiatives. Instead, they rely on methods for transferring value information from existing studies to the policy setting being evaluated. In other words, they develop and apply 'benefit transfer' methods that make best use of the available empirical literature on health preferences. This literature can include QALY-based measures that, like WTP measures, are designed to represent individual preferences.

A critical issue for CBA in public health policy is therefore whether information from QALY studies can be systematically used and combined with WTP studies to develop effective benefit transfer methods. Although primarily designed for CEA, can QALY-based estimates be used to support CBA? This paper uses a specific type of meta-analysis – meta-regression analysis – to address this issue, as it applies to the valuation of avoided acute morbidity.

In several important respects, this study builds on a previous meta-regression analysis conducted by Johnson *et al.* [3]. In particular, we supplement the Johnson *et al.* data with estimates from 12 more recent studies, and we include and control for a number of additional study characteristics through meta-regression. The results of our analysis confirm the findings of Johnson *et al.* with a high level of statistical significance, and they also extend beyond Johnson *et al.* in several ways. In particular, we explicitly test the assumption of a constant WTP per QALY gain. This practice of placing a dollar value on a QALY has been used in several CBA applications [4,5]; however, our results contradict the assumptions underlying this approach.

We begin the next section with a description of meta-regression analysis, showing how it can serve as a tool to synthesize research findings, test hypotheses, and apply for policy analysis. In the following section, we describe a simple conceptual model of health valuation, and in the next section, we define a number of testable hypotheses based on this framework. Then the forthcoming section describes the data used in the analysis, and in the following section, we describe the results of these tests based on meta-regression analyses. Depending on the statistical significance and explanatory power of the meta-regressions, meta-analysis results can also be used for prediction. The ability to reliably predict WTP is an essential criterion for conducting benefit transfer. Therefore, in this context, meta-regression equations can serve as benefit transfer functions. In the penultimate section, we demonstrate and evaluate this application of meta-analysis results. The last section concludes.

Analytical method: meta-regression analysis

Broadly speaking, meta-analysis refers to the practice of using statistical methods to synthesize the results from a well-defined class of empirical studies. Although it has primarily evolved and been applied in the area of health sciences, it is increasingly being used in social science research. Meta-regression analysis, which is a specific kind of meta-analysis, has particularly evolved in economics as a method for summarizing and analyzing empirical results [6].

Smith and Pattanayak [7] provide a recent summary and evaluation of how meta-regression analysis has specifically been used in nonmarket (including health) valuation. They argue that meta-analyses in this field have served three main purposes:

- research synthesis,
- hypothesis testing, and
- prediction.

In this paper, we use meta-regression analysis to address all three of these objectives.

As a method for synthesizing research findings, meta-analysis goes beyond most literature reviews by identifying quantitative measures that can be defined consistently across studies and providing statistical summaries of these measures. In this case, we gather and standardize information on individuals' WTP to avoid acute illness, and we summarize results across a range of studies. The selected studies and WTP estimates for this study are summarized in the 'Data' section and in Appendix A.

Meta-regression analysis expands the scope for hypothesis testing beyond traditional meta-analysis. It allows the analyst to control for and separately estimate the effects of different covariates on the measure of interest. In the context of nonmarket valuation, the core hypothesis of interest is whether observed variation in WTP estimates is consistent with theory. For example, evidence that WTP varies significantly across studies with respect to average income and the size of the quality change [8] supports the validity and credibility of the research.

Conceptual framework for health valuation

To formalize the way in which individuals derive value from changes in health, one can begin with a simple conceptual framework that links an individual's private utility (U) with his or her health status (H). This simple single-period framework, which assumes that H is exogenously determined and that there is no uncertainty regarding H or other determinants of utility, is summarized in the following utility function:

$$U = U(H, X) \tag{1}$$

In this function, X represents a vector of other goods that contribute to utility. Individuals are assumed to maximize this utility function subject to the budget constraint $Y = PX$, where Y is exogenously determined income and P is a vector of prices corresponding to the X goods.

The indirect utility function associated with this maximization process can be written as follows:

$$V = V(H, Y, P) \tag{2}$$

The monetary value, or WTP, associated with an improvement in health from initial health, H^0, to new health, H^1, can be expressed by the compensating variation term (CV) in the following equation:

$$\begin{aligned} V(H^0, Y, P) &= V(H + \Delta H, Y - \mathrm{CV}, P) \\ &= V(H^1, Y - \mathrm{CV}, P) \end{aligned} \tag{3}$$

CV represents the reduction in income that would exactly offset the increase in utility resulting from health improvement, such that there would be no *net* gain in utility. Similarly, the WTP associated with *avoiding* a decline in health from H^0 to H^2 can be expressed by the equivalent variation term (EV) in the following equation:

$$\begin{aligned} V(H^0 - \Delta H, Y, P) &= V(H^2, Y, P) \\ &= V(H^0, Y - \mathrm{EV}, P) \end{aligned} \tag{4}$$

EV represents the reduction in income that would reduce utility by exactly the same amount as the health decline.

Both Equations (3) and (4) can be rearranged to derive the following corresponding value functions:

$$\mathrm{CV} = \mathrm{CV}(H^0, H^1, P, Y) \tag{3$'$}$$

$$\mathrm{EV} = \mathrm{EV}(H^0, H^2, P, Y) \tag{4$'$}$$

These equations represent 'variation' or WTP functions with respect to a change in health status. They provide a conceptual starting point for constructing and estimating meta-analytic functions of WTP for avoided illness. Both CV and EV are considered to be conceptually correct private welfare measures for changes in health; however, even for equivalent gains or losses in health (i.e. holding ΔH constant), they will not necessarily have the same value. The potential differences between these two measures are described in more detail below. CV can also be expressed as the *increase* in income that would exactly offset the utility loss associated with a *decline* in health (i.e. an individual's minimum willingness to accept (WTA) compensation for a decline in health). Although much attention has been devoted in the literature to discrepancies between WTP and WTA measures [9,10], in practice relatively few studies have estimated WTA values for health changes.

Although the CV and EV measures in Equations (3) and (4) are appropriate private welfare measures for health improvements, it is important to emphasize that they will underestimate societal welfare if some costs of illness are externalized to others. As shown by, for example, Harrington and Portney [11], private WTP represented in Equations (3) and (4) should reflect and include individuals' own avoided (1) direct costs of illness, such as medical expenses, (2) indirect costs such as lost income or household production, and (3) pain and suffering. To the extent that portions of these costs are externalized to others, for example through public or private insurance or sick leave policies, they will not be reflected in individuals' private WTP to avoid illness or improve health. These costs, including pain and suffering, may also be externalized through *altruistic* values. That is, certain individuals may lose utility when others experience illness, and they may be willing to pay for the changes in others' health. In particular, parents are often willing to pay for improvements in their children's health. Although it is possible to develop CV and EV measures that incorporate altruistic values for others' health [12,13], their inclusion in CBA is often problematic [14].

Hypotheses regarding determinants of WTP

Based on the logic represented in Equations (3′) and (4′), we expect WTP estimates for health improvements to be primarily influenced by the magnitude of changes in health outcomes and the characteristics of the study population. From an empirical standpoint, we also expect value estimates to be influenced by the valuation method used. Below, we discuss how these determinants of health values can be measured and how they are expected to affect the magnitude of value estimates.

Duration and severity of health effects

Although it is well understood that health outcomes are inherently complex and multidimensional, for the purposes of economic evaluation it is generally necessary to develop simplified summary measures of health. Therefore, we characterize acute conditions in two primary dimensions: duration and severity.

The primary expectation is that larger reductions in the duration and/or severity of illness will generate larger value estimates (positive first derivatives). The test of this type of relationship (i.e. between WTP and the size of the 'commodity' being valued) is commonly referred to as a 'scope test' [8]. The primary challenge in testing for scope effects is therefore to define measures of duration and severity that can be applied consistently across the health effects being valued in the WTP studies.

Specifying a continuous measure of duration for acute conditions is relatively straightforward. For example, it can be expressed as the number of days during which one experiences the acute condition. However, it should be noted that in some instances the acute condition may be experienced as a single event lasting multiple days (e.g. an acute respiratory infection) and in other instances the acute condition may be experienced as multiple discrete events on separate days (e.g. angina or asthma attacks). In both cases, in this paper we use the term duration to refer to the number of days with the acute condition.

Measures of severity, by contrast, are inherently more complex. Fortunately, economists, psychometricians, and other health experts have developed a number of 'generic' health status measures (HSMs) that can be used to characterize severity for a wide range of health outcomes. These measures typically define a set of possible discrete health states, by defining specific health dimensions (e.g. physical, mental, and emotional domains) and discrete levels of health for each dimension. In addition to these methods for categorizing health states, a number of survey-based scoring techniques have been developed for measuring and comparing the severity of health states. These techniques – in particular standard gamble (SG), time tradeoff (TTO), and visual analog scale (VAS) approaches – use preference elicitation methods to produce 'utility weights' for each possible health state (see, for example, [15] for a summary and evaluation of these scoring techniques). Typically these weights are constructed to vary between two extremes – 0 (immediate death) and 1 (perfect health).

Multiattribute utility scales (MAUSs) represent a specific type of HSM. They are of particular interest for quantifying severity of illness because they merge *categorization* of health states (HSM techniques) with *utility measurement* for health states. In other words, they use SG, TTO, or VAS approaches to derive preference-based utility weights/scores (usually on a 0–1 scale) for a wide range of well-defined health states. The three most widely used MAUSs are the Quality of Well-Being (QWB) Scale, the Health Utility Index (HUI), and the EuroQol. Table 1 compares some of the primary features of these three methods (see [16] for a more detailed summary of these methods). Each method has advantages; however, because of the way it specifies dimensions and levels of health, the QWB Scale can be most readily applied as a severity index for acute illness.

One of the distinct features of the QWB Scale is that it provides separate scores for each of the four health dimensions that it covers – symptoms, mobility, physical activity, and social activity. The 27 symptoms range in severity from eye irritation and shortness of breath to loss of consciousness and severe burns. The mobility, physical activity, and social activity scales each primarily include two levels of impairment, one involving modest restrictions and other involving severe restrictions